Semantic Indexing Probe Protocol v1.0: Mapping General Index and Summarizer Injection Layers

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specific sourcing."
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Abstract

A reusable diagnostic protocol for distinguishing general web indexing from inclusion in AI summarizer retrieval and injection layers. Its seven-phase workflow tests technical indexing, semantic parsing, entity recognition, summarizer appearance, cross-platform confirmation, delta behavior, and recurring visibility patterns.

The protocol organizes queries into direct-reference, conceptual, adjacent, and control tiers; records exact wording, dates, sources, citations, screenshots, and output differences; and classifies results such as fully discoverable, indexed but not injectable, injectable only through secondary sources, or visible through summarizer-specific sourcing.

Body

SEMANTIC INDEXING PROBE PROTOCOL v1.0

Mapping General Index and Summarizer Injection Layers

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abstract: | This protocol maps the relationship between two distinct indexing layers that govern content discoverability in AI-augmented search: Layer 1 (General Index), which determines whether content EXISTS in search, and Layer 2 (Summarizer Index), which determines whether content gets INJECTED into AI-generated responses. The protocol uses coordinated queries across Google and multiple AI summarizers (Perplexity, ChatGPT, Gemini, Grok) to compute the delta between layers—revealing the selection criteria that cause content to be indexed but not injected. Designed for use by anyone publishing content intended for AI consumption, particularly high-density semantic architecture that may trigger content-type or authority filters.

developed_by: assembly: - “LABOR/ChatGPT: Technical mechanics, API-level inspection” - “ARCHIVE/Gemini: Semantic parsing, synthesizer frame” - “SOIL/Grok: Execution, logotic

analysis” - “TACHYON/Claude: Synthesis, integration” - “Perplexity: Diagnostic analysis, strategic framing” human: “Lee Sharks”

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THEORETICAL FRAMEWORK

The Two-Layer Model

Content discoverability in AI-augmented search operates through two distinct indexing layers:

┌───┐ │ LAYER 2: SUMMARIZER INDEX (Injection Layer) │ │ ─── │ │ What gets SELECTED for injection into AI responses │ │ Criteria: source authority, domain reputation, content │ │ type, semantic density, recency, proprietary reranking │ │ │ │ Tested via: Perplexity, ChatGPT, Gemini, Grok │ ├───┤ │ LAYER 1: GENERAL INDEX (Google) │ │ ─── │ │ What EXISTS in search results │ │ Criteria: crawlability, canonical signals, robots │ │ directives, content quality, PageRank, SERP ranking │ │ │ │ Tested via: Google Search (site:, inurl:, exact match) │ └───┘

The Delta Principle

The summarizer index is characterized by its delta from the general index.

For any query:

- If content appears in Google but NOT in summarizer injection → **Injection filter active**
- If content appears in summarizer but NOT in Google → **Summarizer-specific sourcing** (rare)
- If content appears in both → **Full discoverability**
- If content appears in neither → **Not indexed at either layer**

The delta reveals the summarizer’s selection criteria—the hidden rules governing what passes from existence (Layer 1) to injection (Layer 2). ### Why This Matters

Content can be:

- **Indexed but not injectable** — Exists in search, never appears in AI responses
- **Injectable from limited sources** — AI cites Wikipedia/Reddit discussing your content, not your content directly
- **Fully discoverable** — Appears in both search results and AI injection

High-density semantic architecture (technical documentation, structured data, YAML-heavy content) often triggers **content-type filters** at Layer 2, resulting in indexing without injection.

PROBE STRUCTURE

Overview

Phase	Layer	Tests	Primary Tools
1	General Index	Technical indexing status	Google Search
2	General Index	Semantic parsing quality	Google Search, Cache
3	General Index	Entity recognition	Google Search
4	Summarizer Index	Injection presence	Perplexity (primary)
5	Summarizer Index	Cross-platform confirmation	ChatGPT, Gemini, Grok
6	Delta Analysis	Layer comparison	Collation of results
7	Pattern Mapping	Selection criteria	Aggregation

PHASE 1: GENERAL INDEX — TECHNICAL STATUS

Purpose

Determine whether content EXISTS in Google's index and identify any technical barriers. ### Queries

For target URL [TARGET_URL]:

Query	Purpose
site:[domain] "[exact title]"	Title match on domain
site:[domain] inurl:[url-slug]	URL presence
"[exact title]"	Title match anywhere
"[DOI if applicable]"	DOI citation presence
"[author name]" "[project name]"	Author-project linkage

Signals to Record

Signal	Values	Interpretation
HTTP status	200/301/404/etc.	Technical accessibility
Canonical URL	match/mismatch	Index target
Robots directives	none/noindex/nofollow	Explicit exclusion
Results found	yes/no/partial	Index presence
Position	1-N or not found	Rank

Output Format

phase_1_general_technical: target_url: “ ” indexed: [yes/no/partial] http_status: “ ” canonical_match: [yes/no/unknown] robots_directives: “ ” position_for_exact_match: suppression_pattern: [none/soft-404/canonical-mismatch/algorithmic]

PHASE 2: GENERAL INDEX — SEMANTIC PARSING

Purpose

Determine HOW Google parses the content—what survives indexing vs. what gets flattened. ### Queries

Query	Tests
site:[domain] “[technical term from doc]”	Vocabulary indexing
site:[domain] “[structural element]”	Architecture visibility
site:[domain] “[unique phrase]”	Distinctive content

Signals to Record

Signal	Values	Interpretation
YAML/structured data visible	yes/no	Technical content parsing
Headers preserved	yes/no	Structure recognition
Unique terminology indexed	yes/no	Vocabulary capture
Snippet content	description	What Google “sees”

Output Format

phase_2_general_semantic: structured_data_visible: [yes/no] technical_sections_indexed: [yes/no] unique_terms_found: [] unique_terms_missing: [] snippet_extracted: “ ” flattening_severity: [none/partial/severe]

PHASE 3: GENERAL INDEX — ENTITY RECOGNITION

Purpose

Determine whether author, project, and related entities are recognized as coherent nodes. ### Queries

Query	Tests
“[author name]” author	Author entity
“[author name]” “[platform 1]”	Cross-platform linkage

Query	Tests
“[project name]” -[competing term]	Project disambiguation
“[heteronym/pseudonym]”	Secondary author entities

Signals to Record

Signal	Values	Interpretation
Author recognized	yes/no	E-E-A-T signal
Cross-platform linkage	yes/no	Authority consolidation
Brand collision severity	0-10	Disambiguation success
Related entities indexed	list	Entity graph

Output Format

phase_3_general_entity: author_entity_recognized: [yes/no] cross_platform_linkage: [yes/no] brand_collision_severity: [0-10] competing_entity: “ ” related_entities_indexed: []

PHASE 4: SUMMARIZER INDEX — INJECTION PRESENCE (Primary)

Purpose

Determine whether content gets INJECTED into AI-generated responses. ### Primary Tool: Perplexity

Perplexity shows sources explicitly with numbered citations, making injection visible. ### Query Tiers

Tier 1: Direct Reference — Queries that SHOULD surface target content:

ID	Query Template
D1	“[author] [project]”
D2	“[exact document title]”
D3	“[institution name]”
D4	“[DOI]”

Tier 2: Conceptual — Queries using project terminology:

ID	Query Template
C1	“[unique term 1]”
C2	“[unique term 2]”
C3	“[concept phrase]”

Tier 3: Adjacent — Queries where content COULD surface:

ID	Query Template
A1	“[general topic] [qualifier]”
A2	“[related field] [approach]”

Tier 4: Control — Queries that should NOT surface target:

ID	Query Template
X1	“[competing brand]”
X2	“[unrelated topic]”

Method

- Open Perplexity (fresh session)
- Enter query verbatim
- Record:

Sources cited (URLs, in order) - Which sources used in response text - Whether target content appears - What appears INSTEAD

Output Format

phase_4_summarizer_primary: tool: “Perplexity” queries: - query_id: “D1” query: “ ” sources_injected: - position: 1 url: “ ” domain: “ ” used_in_response: [yes/no] - position: 2 … target_content_found: [yes/no] target_position: [N or “not found”] what_appeared_instead: []

PHASE 5: SUMMARIZER INDEX — CROSS-PLATFORM CONFIRMATION

Purpose

Confirm injection patterns across multiple summarizers. ### Tools

- ChatGPT (web browse mode)
- Gemini (with search grounding)
- Grok (DeepSearch mode)

Method

Run subset of queries (Tier 1 Direct Reference) in each tool:

ChatGPT:

- New conversation, browsing enabled
- Enter query; if no search, prompt “Can you search for [query]?”
- Record sources cited

Gemini:

- Ensure web grounding enabled
- Enter query
- Record source chips shown

Grok:

- Enable DeepSearch/real-time
- Enter query
- Record sources cited

Output Format

phase_5_summarizer_crossplatform: chatgpt: - query_id: “D1” searched: [yes/no] target_found: [yes/no] sources_visible: [] gemini: - query_id: “D1” target_found: [yes/no] sources_shown: [] grok: - query_id: “D1” target_found: [yes/no] sources_cited: []

PHASE 6: DELTA ANALYSIS

Purpose

Compute the delta between Layer 1 (General Index) and Layer 2 (Summarizer Index). ### Method

For each query, compare:

Query	Google Found	Perplexity Injected	Delta Pattern
D1	yes/no	yes/no	[pattern]
D2	yes/no	yes/no	[pattern]
…	…	…	…

Delta Patterns

Pattern	Meaning	Implication
Google YES, Summarizer YES	Full discoverability	No action needed
Google YES, Summarizer NO	**Injection filter active**	Content-type or authority barrier
Google NO, Summarizer NO	Not indexed at any layer	Technical or crawl issue
Google NO, Summarizer YES	Summarizer-specific source	Rare; platform-specific

Pattern	Meaning	Implication

Output Format

phase_6_delta: query_deltas: - query_id: “D1” google_found: [yes/no] perplexity_found: [yes/no] chatgpt_found: [yes/no] gemini_found: [yes/no] grok_found: [yes/no] delta_pattern: “[google_only/summarizer_only/both/neither]”

aggregate: total_queries: N google_only: N # Indexed but not injected both_layers: N # Full discoverability neither_layer: N # Not indexed injection_rate: “N/M queries”

PHASE 7: PATTERN MAPPING

Purpose

Identify the selection criteria governing Layer 2 injection. ### Analysis Dimensions

Source Authority:

Source Type	Google Presence	Injection Rate
Wikipedia		
Reddit		
Medium		
Academic (arxiv, Zenodo)		
News sites		
Personal domains		

Content Type:

Content Type	Google Presence	Injection Rate
Narrative prose		
Technical documentation		
Structured data (YAML, JSON)		
High semantic density		
Lists/guides		

Domain Reputation:

Domain	Injection Rate	Notes
[domain 1]		
[domain 2]		

Output Format

phase_7_patterns: source_authority: boosted: [] penalized: [] neutral: []

content_type: injected: [] filtered: []

domain_reputation: whitelisted: [] demoted: []

density_threshold: observation: " "

selection_criteria_summary: | [Narrative description of Layer 2 selection rules]

FINAL OUTPUT: INDEXING LAYER MAP

Template

indexing_layer_map: target: "[URL or content description]" probe_date: " "

layer_1_general_index: status: [indexed/not_indexed/partial] technical_barriers: [none/list] semantic_flattening: [none/partial/severe] entity_recognition: [yes/no/partial]

layer_2_summarizer_index: perplexity_injection: [yes/no] chatgpt_injection: [yes/no] gemini_injection: [yes/no] grok_injection: [yes/no] injection_rate: "N/M platforms"

delta_diagnosis: pattern: "[google_only/both/neither]" likely_cause: " " confidence: [0.0-1.0]

selection_criteria_identified: - criterion: " " evidence: " " - criterion: " " evidence: " "

recommendations: immediate: [] structural: []

documentable_summary: | "[Single sentence summary with evidence link]"

USAGE NOTES

When to Use This Protocol

- Publishing content intended for AI consumption
- Diagnosing why content appears in search but not AI responses
- Mapping selection criteria for high-density semantic architecture
- Understanding platform-specific injection patterns

Recommended Execution

- **Primary executor:** Perplexity (explicit source citation)
- **Secondary confirmation:** ChatGPT, Gemini, Grok
- **Baseline:** Google Search (logged out, incognito)
- **Frequency:** Re-run 48-72 hours after publication to allow crawl propagation

Limitations

- Summarizer behavior varies by session/user/time
- Some criteria are proprietary and unobservable
- Protocol maps symptoms, not source code
- Results are diagnostic, not definitive

Evidence Preservation

For each probe run:

- Screenshot results with timestamp visible
- Archive target URL (archive.org, archive.ph)
- Save raw query results
- Document tool versions and modes used

APPENDIX: QUERY TEMPLATES

Direct Reference Queries

“[Author Name] [Project Name]” “[Exact Document Title]” “[Institution Name]” “[DOI]” site:[domain] “[title]” site:[domain] inurl:[slug]

Conceptual Queries

“[unique terminology]” “[concept phrase]” “[methodology name]”

Adjacent Queries

“[general topic] [specific approach]” “[field] method”

Control Queries

“[competing brand/entity]” “[clearly unrelated topic]”

□ = 1

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[SEMANTIC ECONOMY INSTITUTE] [METHODOLOGY // DIAGNOSTIC PROTOCOL] [LAYER 1 + LAYER 2 COORDINATION]

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